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Inebriation test system

Abstract

A method to prevent intoxicated operation is described. The method includes providing instructions to a vehicle user to position a vehicle user body portion posture in a predefined alignment. The method further includes obtaining the vehicle user body portion posture from a first vehicle detector, and determining whether the vehicle user body portion posture is in the predefined alignment. The method includes activating a plurality of vehicle visual indicators to illuminate in a predefined manner when the vehicle user body portion posture is in the predefined alignment. The method further includes providing instructions to the vehicle user to move vehicle user eyes to track the plurality of vehicle visual indicators, and obtaining a vehicle user eye movement from a second vehicle detector. The method further includes determining whether the vehicle user eye movement meets a predetermined condition, and actuating a control action accordingly.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8226574	12/2011	Whillock	351/210	A61B 5/4863
9475387	12/2015	Wu	N/A	N/A
10089543	12/2017	Fung	N/A	G06V 40/168
10343693	12/2018	Plummer et al.	N/A	N/A
10390802	12/2018	Picard	N/A	A61B 3/113
2007/0132950	12/2006	Victor	351/200	A61B 3/1015
2010/0280372	12/2009	Poolman	600/407	A61B 5/4094
2010/0294583	12/2009	Biondo	340/576	A61B 5/082
2015/0158495	12/2014	Duncan	701/1	G08G 1/163
2015/0258892	12/2014	Wu	340/576	G06V 40/166
2016/0339922	12/2015	Schmidt	N/A	B60W 40/08
2017/0049362	12/2016	Macknik	N/A	A61B 5/163
2017/0105669	12/2016	Valenti	N/A	A61B 3/0025
2018/0301225	12/2017	Nothacker et al.	N/A	N/A
2018/0339706	12/2017	Biondo	N/A	B60W 50/10
2021/0200991	12/2020	Prins	N/A	G06V 40/19
2021/0291650	12/2020	Minjeur	N/A	A61B 5/0002
2021/0316737	12/2020	Iwase	N/A	A61B 5/4035
2021/0394764	12/2020	Clines	N/A	G06F 16/93
2022/0153302	12/2021	Arechiga-Gonzalez	N/A	B60W 50/14
2023/0233120	12/2022	Fichtler	600/301	A61B 5/7264

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102785574	12/2014	CN	N/A
102975721	12/2015	CN	N/A

OTHER PUBLICATIONS

Guoliang Yuan et al., A Novel Driving Behavior Learning and Visualization Method with Natural Gaze Prediction, IEEE Access, vol. 9, Jan. 27, 2021, 18560-18568. cited by applicant
Mnay K Diddi1 et al., Head Pose and Eye State Monitoring (HEM) for Driver Drowsiness Detection: Overview, JISET—International Journal of Innovative Science, Engineering & Technology, vol. 1, Issue 9, Nov. 9, 2014, 504-508. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to mitigating incidents of intoxicated driving, and more particularly, for vehicle systems that test for driver inebriation using eye gaze.

BACKGROUND

(2) Government and private entities frequently implement initiatives that prevent intoxicated operation of motor vehicles. For example, state and federal agencies may provide driver awareness programs. Some corporations educate employees on driver inebriation consequences, and schools and colleges frequently teach their students negative outcomes associated with intoxicated vehicle operation. Even with these initiatives, some vehicle drivers may still choose to operate vehicles while intoxicated despite efforts to minimize such instances.

(3) Authorities implement various approaches to detect driver inebriation. One conventional approach includes the use of a breath analyzer. The breath analyzer measures an alcohol amount in a person's breath and helps to minimize intoxicated operation of vehicles. Typically, enforcement officers, e.g., police officers, use breath analyzers to test vehicle drivers' breath, and prevent the drivers from driving their vehicles if the breath indicates alcohol presence. This approach of preventing intoxicated operation is effective; however, implementation of education and enforcement measures are limited. In addition, a separate apparatus, for example, the breath analyzer, is required to test the drivers' breath. Therefore, this approach may not always be practical, and may be limited in its utility due to logistical and/or bandwidth constraints.

(4) Another conventional approach is the use of a field sobriety test that includes a horizontal gaze nystagmus test. To take the horizontal gaze nystagmus test, a police officer may ask the vehicle driver to track movement of an object, such as a flashlight or a pen, from side to side. As the vehicle driver tracks the object movement and rotates his eyes, eye twitching may occur at extreme eye poses, and/or eye movement may become irregular. The police officer may detect driver inebriation when eye twitching and/or irregular eye movement occurs. Although this approach does not require a separate apparatus (e.g., the breath analyzer), this approach too is limited by police officers' bandwidth/availability.

(5) Thus, there still exists a need in the industry for a system and method for preventing intoxicated

operation, specifically for determining driver inebriation and executing control actions.
(6) It is with respect to these and other considerations that the disclosure made herein is presented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The detailed description is set forth with reference to the accompanying drawings. The use of the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.
- (2) FIG. 1 depicts an example system to prevent intoxicated operation of a vehicle, in accordance with the present disclosure.
- (3) FIG. 2 depicts an example vehicle in which techniques and structures for providing the systems and methods disclosed herein may be implemented.
- (4) FIG. 3 illustrates an example embodiment of providing training to a vehicle user to take an inebriation test in a vehicle in accordance with the present disclosure.
- (5) FIG. 4 illustrates an example control action(s) for mitigating intoxicated operation in accordance with the present disclosure.
- (6) FIG. 5 depicts a first flow diagram of an example method for providing training to a vehicle user to take an inebriation test, in accordance with the present disclosure.
- (7) FIG. 6 depicts a second flow diagram of an example method for providing training to a vehicle user to take an inebriation test, in accordance with the present disclosure.

DETAILED DESCRIPTION

(8) Overview

- (9) The present disclosure describes a system and method to reduce or prevent instances of vehicle operation while intoxicated, whether through consumption of alcohol or some other substance that would impair one's ability to safely operate a vehicle. In particular, the present disclosure describes a system configured and/or programmed to coach a vehicle user to perform an inebriation test (such as a horizontal eye gaze nystagmus test) inside a vehicle. In some aspects, a vehicle processor may provide an instruction (also referred to as output or feedback) to the vehicle user to position user's head in a predetermined alignment, for example, a straight head position aligned towards a vehicle steering wheel.
- (10) Responsive to the vehicle user positioning her/his head in the predetermined alignment, the vehicle processor may activate a plurality of vehicle visual Human-Machine Interface (HMI) components in a predefined manner. For example, the vehicle processor may activate the visual HMI components in a sequential manner, from one vehicle end (e.g., a left vehicle end) to another (e.g., a right vehicle end).
- (11) In some aspects, the vehicle processor may provide another instruction to the vehicle user to track the plurality of vehicle visual HMI components by using vehicle user eyes, without moving the head. The vehicle processor may determine whether a vehicle user eye movement (during the tracking) meets a predetermined condition and may actuate one or more control action(s). In some aspects, the vehicle processor may determine that the vehicle user is in an inebriated state when the vehicle user eye movement meets the predetermined condition.
- (12) The predetermined condition can include eye twitching at extreme eye gazes or an irregular eye tracking when the vehicle user eyes track the plurality of vehicle visual HMI components.
- (13) In some aspects, responsive to a determination that the vehicle user is in the inebriated state, the vehicle processor may actuate the control action(s). The control action(s) may include disabling

a vehicle start, transmitting a vehicle user inebriated state notification to an acquaintance user device or a vehicle ridesharing server, providing an instruction to a non-inebriated vehicle passenger to drive the vehicle, and/or calling authorities (for example, the police).

(14) In some aspects, the vehicle processor may trigger the inebriation test when the vehicle user enters the vehicle or based on trigger events. The trigger events may include, for example, indication of distracted driving or other vehicle user behaviors. For instance, if the vehicle user is swerving, feeling drowsy or fatigued, the vehicle processor may trigger the inebriation test. In some aspects, the vehicle user may trigger the inebriation test to confirm if the vehicle user is fit for driving. In further aspects, the inebriation test may be triggered automatically based upon user settings.

(15) The present disclosure provides various advantages over the conventional approaches to mitigate or prevent intoxicated operation of a motor vehicle configured with the disclosed system. The vehicle may provide coaching to the vehicle user to take the inebriation test without involvement of any third party (such as enforcement officers or police officers). Further, the system does not require addition of a breath analyzer to detect a vehicle user inebriation state.

(16) These and other advantages of the present disclosure are provided in detail herein.

Illustrative Embodiments

(17) The disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the disclosure are shown, and not intended to be limiting.

(18) FIG. 1 depicts a system **100** to prevent intoxicated operation, in accordance with the present disclosure. More particularly, FIG. 1 depicts the system **100** that may include a vehicle **102**. The vehicle **102** may take the form of any passenger or commercial vehicle such as, for example, a car, a work vehicle, a crossover vehicle, a van, a minivan, a taxi, a bus, etc. In addition, the vehicle **102** may include other land vehicles (such as motorcycles, scooters etc.), marine vehicles (such as water scooter, jet skis, boats etc.) and/or the like. Further, the vehicle **102** may be a manually driven vehicle, and/or may be configured to operate in a partially autonomous mode, and may include any powertrain such as, for example, a gasoline engine, one or more electrically actuated motor(s), a hybrid system, etc.

(19) In some aspects, the vehicle **102** may include an automotive computer **104**, a Vehicle Control Unit (VCU) **106**, and an intoxicated operation prevention unit **110**. The VCU **106** may include a plurality of Electronic Control Units (ECUs) **108** disposed in communication with the automotive computer **104**.

(20) The system **100** may further include a mobile device **112** that may connect with the automotive computer **104** and/or the intoxicated operation prevention unit **110** by using wired and/or wireless communication protocols and transceivers. In some aspects, the mobile device **112** may be associated with a vehicle user/operator (not shown in FIG. 1). In other aspects, the mobile device **112** may be associated with other users (not shown in FIG. 1). The mobile device **112** may communicatively couple with the vehicle **102** via one or more network(s) **114**, which may communicate via one or more wireless connection(s), and/or may connect with the vehicle **102** directly by using near field communication (NFC) protocols, Bluetooth® protocols, Wi-Fi, Ultra-Wide Band (UWB), and other possible data connection and sharing techniques.

(21) The network(s) **114** illustrates an example communication infrastructure in which the connected devices discussed in various embodiments of this disclosure may communicate. The network(s) **114** may be and/or include the Internet, a private network, public network or other configuration that operates using any one or more known communication protocols such as, for example, transmission control protocol/Internet protocol (TCP/IP), Bluetooth®, Bluetooth® Low Energy (BLE), Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) standard 802.11, UWB, and cellular technologies such as Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA), High-Speed Packet Access (HSPDA), Long-Term Evolution

(LTE), Global System for Mobile Communications (GSM), and Fifth Generation (5G), to name a few examples.

(22) In some aspects, the automotive computer **104** and/or the intoxicated operation prevention unit **110** may be installed in a vehicle engine compartment (or elsewhere in the vehicle **102**) and operate as an intoxicated operation prevention unit functional part, in accordance with the disclosure. The automotive computer **104** may be or include an electronic vehicle controller, having one or more processor(s) **116** and a memory **118**. Moreover, the intoxicated operation prevention unit **110** may be separate from the automotive computer **104** (as shown in FIG. **1**) or may be integrated as an automotive computer part.

(23) The processor(s) **116** may be disposed in communication with one or more memory devices disposed in communication with the respective computing systems (e.g., the memory **118** and/or one or more external databases not shown in FIG. **1**). The processor(s) **116** may utilize the memory **118** to store programs in code and/or to store data for performing aspects in accordance with the disclosure. The memory **118** may be a non-transitory computer-readable memory storing a vehicle intoxicated operation prevention program code. The memory **118** can include any one or a combination of volatile memory elements (e.g., dynamic random-access memory (DRAM), synchronous dynamic random-access memory (SDRAM), etc.) and can include any one or more nonvolatile memory elements (e.g., erasable programmable read-only memory (EPROM), flash memory, electronically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), etc.).

(24) In some aspects, the automotive computer **104** may be disposed in communication with one or more server(s) **120**, and the mobile device **112**. The server(s) **120** may be part of a cloud-based computing infrastructure and may be associated with and/or include a Telematics Service Delivery Network (SDN) that provides digital data services to the vehicle **102** and other vehicles (not shown in FIG. **1**) that may be part of a vehicle fleet.

(25) In accordance with some aspects, the VCU **106** may share a power bus with the automotive computer **104** and may be configured and/or programmed to coordinate the data between vehicle **102** systems, connected servers (e.g., the server(s) **120**), and other vehicles (not shown in FIG. **1**) operating as part of a vehicle fleet. The VCU **106** can include or communicate with any combination of the ECUs **108**, such as, for example, a Body Control Module (BCM) **122**, an Engine Control Module (ECM) **124**, a Transmission Control Module (TCM) **126**, a telematics control unit (TCU) **128**, a Driver Assistances Technologies (DAT) controller **130**, etc. The VCU **106** may further include and/or communicate with a Vehicle Perception System (VPS) **132**, having connectivity with and/or control of one or more vehicle sensory system(s) **134**. In some aspects, the VCU **106** may control the vehicle **102** operational aspects and implement one or more instruction sets received from the application operating on the mobile device **112**, from one or more instruction sets stored in the memory **118**, including instruction sets stored in the intoxicated operation prevention unit **110**.

(26) The TCU **128** can be configured and/or programmed to provide vehicle connectivity to wireless computing systems onboard and off board the vehicle **102**, and may include a Navigation (NAV) receiver **136** for receiving and processing a GPS signal, a BLE Module (BLEM) **138**, a Wi-Fi transceiver, a UWB transceiver, a NFC transceiver and/or other wireless transceivers (not shown in FIG. **1**) that may be configurable for wireless communication between the vehicle **102** and other systems, computers, and modules. The TCU **128** may be disposed in communication with the ECUs **108** by way of a bus.

(27) In some aspects, the ECUs **108** may control aspects of vehicle operation and communication using inputs from human drivers, inputs from an autonomous vehicle controller, the intoxicated operation prevention unit **110**, and/or via wireless signal inputs received via the wireless connection(s) from other connected devices, such as the mobile device **112**, the server(s) **120**, among others.

(28) The BCM **122** generally includes integration of sensors, vehicle performance indicators, and variable reactors associated with vehicle systems, and may include processor-based power distribution circuitry that can control functions associated with the vehicle body such as lights, windows, security, camera(s), audio system(s), speakers, display system, door locks and access control, vehicle energy management, and various comfort controls. The BCM **122** may also operate as a gateway for bus and network interfaces to interact with remote ECUs (not shown in FIG. **1**).

(29) In some aspects, the DAT controller **130** may provide Level-1 through Level-3 automated driving and driver assistance functionality that can include, for example, active parking assistance, trailer backup assistance, adaptive cruise control, lane keeping, and/or driver status monitoring, among other features. The DAT controller **130** may also provide aspects of user and environmental inputs usable for user authentication.

(30) The DAT controller **130** can further obtain input information via the vehicle sensory system(s) **134**, which may include sensors disposed on the vehicle interior and/or exterior (sensors not shown in FIG. **1**). The DAT controller **130** may receive the sensor information associated with driver functions, vehicle functions, and environmental inputs, and other information.

(31) In some aspects, the automotive computer **104** may connect with an infotainment system **152** that may include a touchscreen interface portion, and may include voice recognition features, biometric identification capabilities that can identify users based on facial recognition, voice recognition, fingerprint identification, or other biological identification means. In other aspects, the infotainment system **152** may provide user identification using mobile device pairing techniques (e.g., connecting with the mobile device **212**, a Personal Identification Number (PIN) code, a password, passphrase, or other identifying means).

(32) The computing system architecture of the automotive computer **104**, the VCU **106**, and/or the intoxicated operation prevention unit **110** may omit certain computing modules. It should be readily understood that the computing environment depicted in FIG. **1** is an example of a possible implementation according to the present disclosure, and thus, it should not be considered limiting or exclusive.

(33) In accordance with some aspects, the intoxicated operation prevention unit **110** may be executed as part of and/or integrated with the ECUs **108**. The intoxicated operation prevention unit **110**, regardless of whether it is integrated with the automotive computer **104** or the ECUs **108**, or whether it operates as an independent computing system in the vehicle **102**, may include a detection unit **140** (or detector(s) **140**), a processor **142**, and a computer-readable memory **144**, which are communicatively coupled to each other. The detector(s) **140** may include a first detector **154** and a second detector **156**.

(34) In accordance with some aspects, the intoxicated operation prevention unit **110** may be configured to take an inebriation test of a vehicle user, for example a vehicle **102** driver (not shown in FIG. **1**). In particular, the intoxicated operation prevention unit **110** may be configured to provide training to the vehicle user to take the inebriation test inside the vehicle **102** and actuate control actions accordingly. The details of the above-mentioned features may be understood as follows.

(35) In some aspects, the intoxicated operation prevention unit **110** may actuate, via the processor **142**, the vehicle user inebriation test when the vehicle user enters the vehicle **102** and/or when the processor **142** receives a trigger event, as discussed below.

(36) In some aspects, the intoxicated operation prevention unit **110** may actuate the inebriation test whenever the vehicle user enters the vehicle **102** if the vehicle user has a history of intoxicated operation. In particular, the memory **118** and/or the memory **144** may store a vehicle user intoxicated operation history, and the processor **142** may fetch the history when the vehicle user enters the vehicle **102**. In addition, or alternatively, the processor **142** may obtain the vehicle user intoxicated operation history from the server **120** (which may, for example, store intoxicated operation histories of a plurality of vehicle users). Responsive to obtaining the intoxicated operation history, the processor **142** may actuate the vehicle user inebriation test if the history

shows repeated intoxicated operation instances.

(37) In some aspects, the intoxicated operation prevention unit **110** may determine the vehicle user entry in the vehicle **102** via the sensory system **134**. For example, the sensory system **134** may include vehicle occupancy sensors that may send a signal to the processor **142**, when the vehicle user enters the vehicle **102**. Responsive to receiving the vehicle user entry signal, the processor **142** may actuate the vehicle user inebriation test, as described above.

(38) In another aspect, the intoxicated operation prevention unit **110** may actuate a general vehicle user focus test when the vehicle user enters the vehicle **102**, before actuating the inebriation test. For example, the processor **142** may actuate the general vehicle user focus test, when the vehicle user does not have an intoxicated operation history. In this case, the processor **142** may instruct the vehicle user to take the general vehicle user focus test, and based on the focus test results, the processor **142** may actuate (or not actuate) the inebriation test. In one aspect, the processor **142** may actuate the inebriation test, when the vehicle user does not pass the general vehicle user focus test. The general vehicle user focus test details are described in conjunction with FIG. 3.

(39) In another aspect, the processor **142** may actuate the vehicle user inebriation test when the processor **142** determines that the vehicle user is swerving while driving the vehicle **102**. In particular, the processor **142** may determine whether the vehicle user is swerving the vehicle **102** via the DAT controller **130**. For example, the DAT controller **130** may determine that the vehicle user is changing lanes frequently (e.g., above a predefined threshold in a preset period) and may send a signal associated with the frequency of lane changing to the processor **142**. Responsive to receiving the lane-changing signal, the processor **142** may instruct the vehicle user to pull over and may actuate the vehicle user inebriation test.

(40) In yet another aspect, the processor **142** may actuate the vehicle user inebriation test when the processor **142** determines that the vehicle user may be fatigued. In some aspects, the processor **142** may determine that the vehicle user may be fatigued by measuring vehicle user eye closure percentage, when the vehicle user is driving the vehicle **102** or otherwise. For example, a vehicle camera (e.g., a vehicle driver tracking camera or a vehicle occupant camera) may capture vehicle user eyes and may send the captured eye information to the processor **142**. Responsive to receiving the eye information, the processor **142** may determine (e.g., by using deep or convolutional neural network architecture and/or training set) whether the vehicle user eye closure percentage is greater than a predetermined threshold. Responsive to determining that the closure percentage is greater than the predetermined threshold, the processor **142** may determine that the vehicle user may be fatigued and may actuate the inebriation test (or may actuate the general vehicle user focus test before actuating the inebriation test).

(41) In further aspects, the processor **142** may determine whether the vehicle user may be fatigued by evaluating a Vestibulo-Ocular Reflex (VOR) of the vehicle user which may be caused in response to vehicle movement (e.g., when the vehicle user drives the vehicle **102**). Specifically, the processor **142** may evaluate vehicle user's VOR performance to determine whether the vehicle user may be fatigued.

(42) In other aspects, the processor **142** may determine whether the vehicle user may be fatigued by measuring additional vehicle user parameters. For example, the processor **142** may determine a vehicle user engagement level with the vehicle steering wheel (not shown in FIG. 1), to determine whether the vehicle user is fatigued. In particular, the processor **142** may determine, via a vehicle user steering wheel sensor (not shown in FIG. 1) if the vehicle user is holding or engaging with the steering wheel too loosely or too tightly. Based on the determination of an abnormal engagement level, the processor **142** may determine that the vehicle user may be fatigued and may actuate the inebriation test (or may actuate the general vehicle user focus test before actuating the inebriation test).

(43) In some aspects, the vehicle user may request the processor **142** to actuate the inebriation test. For example, the vehicle user may send a request, via a vehicle user device or the infotainment

system **152**, to the processor **142** to actuate the inebriation test, if the vehicle user wants to check whether he is fit to drive the vehicle **102**. Responsive to receiving the request from the vehicle user, the processor **142** may actuate the inebriation test.

(44) Responsive to the inebriation test actuation, the first detector **154** may detect a vehicle user's body portion inside the vehicle **102**. In particular, the first detector **154** may be configured to detect a vehicle user's head posture. In one or more aspects, the first detector **154** may include a vehicle interior camera(s), a vehicle driver tracking camera, a vehicle occupant camera, and/or any other vehicle component configurable to detect the vehicle user's head posture. In some aspects, the first detector **154** may monitor/track the vehicle user's head posture regularly and/or at predetermined intervals, when the processor **142** actuates the inebriation test.

(45) In some aspects, the processor **142** may be configured to receive the vehicle user head posture information from the first detector **154**. Responsive to receiving the head posture information, the processor **142** may store the information in the memory **144**. In particular, the processor **142** may store the head posture information in a vehicle user head posture (UHP) database **146**. In addition, the processor **142** may determine whether the head posture is in a predefined alignment, based on the received information. In some aspects, the predefined alignment may include the head posture being straight (and not oriented sideways, upwards or downwards).

(46) In a scenario where the processor **142** determines that the head posture is in the predefined alignment, the processor **142** may initiate a vehicle user eye movement test (e.g., a horizontal eye gaze nystagmus test), described below. On the other hand, if the processor **142** determines that the head posture is not in the predefined alignment (e.g., tilted right or left), the processor **142** may provide feedback to the vehicle user to correct the head posture. For instance, the processor **142** may instruct the vehicle user, via the infotainment system **152**, to look straight, upwards (or downwards), and/or the like. In some aspects, the processor **142** may instruct the user to correct the head posture alignment before starting the horizontal eye gaze nystagmus test and/or during the test. For example, the processor **142** may instruct the vehicle user to keep the head posture fixed in a straight alignment, during the horizontal eye gaze nystagmus test duration. In some aspects, the processor **142** may provide multiple feedbacks/instructions to the vehicle user, until the head posture is in the predefined alignment. The details of head posture correction may be understood in conjunction with FIG. 2.

(47) A person ordinarily skilled in the art may appreciate that the processor **142** corrects the vehicle user head posture to perform the horizontal eye gaze nystagmus test accurately. Specifically, the horizontal eye gaze nystagmus test may involve determination of vehicle user's eye twitching or irregular eye movement, as the vehicle user rotates eyeballs in a horizontal direction (e.g., left or right) during the test. If the vehicle user's head posture is not aligned correctly, the processor **142** may not be able to determine the eye twitching and/or irregular eye movement. Therefore, the processor **142** may prompt the vehicle user to correct the vehicle user's head posture by providing instructions. In some aspects, the processor **142** may display the vehicle user's head posture on the infotainment system **152** and may further display the target posture (i.e., the predefined alignment) on the infotainment system **152**, e.g., as a mannequin head. The processor **142** may wait until the vehicle user's head posture is correct. Therefore, before initiating the horizontal eye gaze nystagmus test, the processor **142** may prompt the user to correct the vehicle user's head posture, till the user corrects the head posture. In other aspects, the vehicle **102** may include a robotic arm that may physically correct the user's head posture.

(48) Responsive to a determination that the vehicle user's head posture is in the predefined alignment, the processor **142** may initiate the horizontal eye gaze nystagmus test. In some aspects, to initiate the horizontal eye gaze nystagmus test, the processor **142** may activate a plurality of vehicle visual indicators to illuminate in a predetermined manner, and may instruct the vehicle user, via the infotainment system **152**, to track the activated indicators. In particular, the processor **142** may activate the plurality of vehicle visual indicators in a sequential manner such that the vehicle

user rotates the eyes in a horizontal direction, to track the activated indicators. In some aspects, the plurality of vehicle visual indicators may include Human-Machine Interface (HMI) visual components (or vehicle key-point illuminators). For instance, the visual indicators may be ambient vehicle lights, mirror blind spot indicators, overhead lamps, and/or the like. Alternatively, or additionally, the processor **142** may activate a vehicle display(s), for example a vehicle panoramic display, and may instruct the vehicle user to track the activated display. The details of vehicle visual indicator activation may be understood in conjunction with FIG. **3**. In addition, the processor **142** may provide the instructions to the vehicle user **202** to track the activated indicators in the form of audio feedback (e.g., spoken text commands, via a vehicle audio system or the infotainment system **152**).

(49) In some aspects, the processor **142** may command the second detector **156** to detect or capture the vehicle user's eye movement, as the vehicle user tracks the activated indicators. The second detector **156** may include, for example, an eye gazing device or any other vehicle component (e.g., the vehicle driver tracking camera or the vehicle occupant camera) configurable to detect the vehicle user's eye movement. In some aspects, the second detector **156** may be same as the first detector **154**. In other aspects, the second detector **156** may be different from the first detector **154**.

(50) In some aspects, the processor **142** may receive the vehicle user's eye movement information from the second detector **156** and may store the received information in a vehicle user eye movement (UEM) database **148**. In addition, the processor **142** may determine user's eye twitching at extreme eye gazes, irregular eye movement or tracking, and/or evaluate pupil or glint trajectory throughout the testing process, based on the received eye movement information. In some aspects, the processor **142** may determine the user's eye movement by localizing the eyes using regression or other classification approaches.

(51) Specifically, responsive to receiving the eye movement information, the processor **142** may run a single deep-learning algorithm that does multi-frame analysis, a pair of deep-learning algorithms to detect eye twitching and irregular eye movement/tracking, or a series of deterministic algorithms that evaluate pupil trajectory throughout the testing process. In some aspects, the algorithms may be deep or convolutional neural network architecture-based algorithms. For instance, the algorithms may include long short-term memory (LSTM) neural network based algorithms, recurrent neural network (RNN) based algorithms, spatio-temporo neural network based algorithms and/or the like.

(52) In particular, the processor **142** may be configured to determine, by using deep or convolutional neural network-based algorithms, whether the vehicle user's eye movement meets a predetermined condition, based on the received eye movement information. In some aspects, the predetermined condition may be associated with a vehicle user's inebriated state. In particular, the predetermined condition may include eye twitching at extreme eye gazes or an irregular eye movement, when the vehicle user's eyes track the activated indicators. A person ordinarily skilled in the art may appreciate that the vehicle user's eyes may twitch at extreme gazes when the vehicle user is in an inebriated state.

(53) In accordance with some aspects, the processor **142** may be configured to actuate a control action(s), responsive to the determination that the vehicle user's eye movement meets the predetermined condition. The control actions may include, for example, calling a vehicle user's friend or family, calling police, a vehicle sharing service, and/or the like. The control action details may be understood in conjunction with FIG. **4**.

(54) The processor **142** may fetch one or more instruction sets having instructions for control actions, and retrieve user profile information (user profile information not shown in FIG. **1**) from the memory **144**. Specifically, the memory **144** may store a vehicle user profile information in a user information (UI) database **150**. The information may include, for example, vehicle username, address, contact list, information associated with user's daily activities, user's intoxicated operation history, and/or the like. In some aspects, the processor **142** may access the vehicle user's contact list

to contact a friend, when the processor **142** determines that the vehicle user is in the inebriated state.

(55) FIG. 2 depicts an example vehicle **200** in which techniques and structures for providing the systems and methods disclosed herein may be implemented. The vehicle **200** may be the same as the vehicle **102**. A vehicle user **202** (or a driver **202**) may operate the vehicle **200**. As discussed above, the first detector **154** may be configured to monitor a vehicle user's head posture, when the vehicle user **202** is sitting inside the vehicle **200**. In some aspects, the first detector **154** may monitor the vehicle user head posture when the processor **142** initiates the inebriation test. In some aspects, the first detector **154** may monitor the vehicle user head posture throughout the inebriation test process.

(56) As described above, the first detector **154** may send the vehicle user head posture information to the processor **142**, which may determine whether the head posture is in the predefined alignment. Responsive to a determination that the head posture is not in the predefined alignment, the processor **142** may instruct the vehicle user to correct the head posture.

(57) For instance, if the processor **142** determines that the vehicle user head is rotated left, as shown in FIG. 2A, the processor **142** may instruct the vehicle user **202** to correct the head posture and “look straight” (e.g., in the predefined alignment). In some aspects, the processor **142** may provide the instructions to the vehicle user **202** to correct the head posture in the form of audio or video feedback (e.g., alarms or spoken text commands, via a vehicle audio system or an infotainment system **204**). For example, the processor **142** may instruct the vehicle user to “rotate head to the right” or “align head with the center console” or “tilt head more towards the steering wheel”, to correct the vehicle user head posture. In addition, the feedback can take the form of directional LED lights or information on the infotainment system **204** including text, arrow indicators, or a figure of a head that moves with the user's head that is displayed on the infotainment system **204**.

(58) Similarly, if the vehicle user head is rotated slightly rightwards (as shown in FIG. 2C) or towards an extreme right position (as shown in FIG. 2D), the processor **142** may instruct the vehicle user **202** to correct the head posture and may provide audio/video feedback according to the user head tilt or rotation.

(59) In some aspects, the processor **142** may initiate the horizontal eye gaze nystagmus test, when the vehicle user **202** corrects the head posture to align straight, as shown in FIG. 2B. The process of conducting the horizontal eye gaze nystagmus test, via the processor **142**, may be understood in conjunction with FIG. 3.

(60) In some aspects, the first detector **154** may keep on monitoring the vehicle user head, even when the processor **142** initiates the horizontal eye gaze nystagmus test. In other words, even during the horizontal eye gaze nystagmus test, if the vehicle user **202** changes head position from “A” to “B”, “C” or “D”, the processor **142** may provide real-time instructions or feedback to the vehicle user **202** to keep the head posture fixed at position “A”. This is because if the vehicle user **202** moves the head (instead of eyes) to track the visual indicators, then the processor **142** may not be able to detect the eye twitching correctly, as the vehicle user **202** may not be rotating his eyes in the required manner (and may instead be rotating his head). Hence, the inebriation test results may not be accurate if the vehicle user **202** moves the head.

(61) In some aspects, the processor **142** may inform the vehicle user **202**, via the infotainment system **204**, that the vehicle user **202** does not pass the inebriation test, if the vehicle user **202** does not correct the head posture after a set number of feedback or instructions. In this case, the processor **142** may re-initiate the horizontal eye gaze nystagmus test or may determine that the vehicle user **202** is in an inebriation state (and take control actions, as described in FIG. 4). In some aspects, the processor **142** may re-initiate the horizontal eye gaze nystagmus test if the vehicle user **202** does not correct the head posture after receiving a first predefined number of feedback/instructions. Similarly, the processor **142** may determine that the vehicle user **202** is in the

inebriation state if the vehicle user **202** does not correct the head posture after receiving a second predefined number of feedback/instructions. In some aspects, the second predefined number of feedback/instructions may be greater than the first predefined number of feedback/instructions.

(62) FIG. **3** illustrates an example embodiment of providing training to a vehicle user **302** to take an inebriation test in a vehicle **300** in accordance with the present disclosure. The vehicle **300** may be same as the vehicle **102**. In some aspects, the vehicle **300** may be configured to guide or coach the vehicle user **302** to take the inebriation test (such as the horizontal eye gaze nystagmus test) inside the vehicle **300**.

(63) In particular, the vehicle **300** may be configured to conduct the inebriation test by using a plurality of vehicle visual indicators. The vehicle visual indicators may include Human-Machine Interface (HMI) visual components (or vehicle key-point illuminators). For instance, the visual indicators may be ambient vehicle lights, mirror blind spot indicators, overhead lamps, and/or the like. Alternatively, or additionally, the vehicle **300** may be configured to conduct the test by using vehicle displays, such as one or more vehicle panoramic displays (not shown in FIG. **3**), a heads-up display (HUD) (not shown in FIG. **3**), or another user interface.

(64) As discussed above, the processor **142** may activate the inebriation test when the vehicle user **302** enters the vehicle **300** and has an intoxicated operation history (based on, for example, the user profile history) when the processor **142** determines one or more predefined trigger events (e.g., when the vehicle user **302** is swerving the vehicle **300** or when the vehicle user **302** is fatigued), or when the vehicle user **302** requests inebriation test actuation.

(65) Responsive to the inebriation test activation, the processor **142** may confirm the vehicle user head posture, as described in FIGS. **1** and **2**. When the head posture is in the predetermined alignment, the processor **142** may activate the vehicle visual indicators (or vehicle display) in a predetermined manner (e.g., from one side to another to capture irregular eye twitching at extreme eye poses). The processor **142** may further instruct the vehicle user **302** to track the vehicle visual indicators without moving the head. In some aspects, the processor **142** may provide the instructions to the vehicle user **302** in the form of audio or video feedback (such as via a vehicle audio system or an infotainment system **304**). In some aspects, the processor **142** may provide the instructions to the vehicle user **302** via a vehicle user device (not shown in FIG. **3**) that may operatively connect with the vehicle **300**.

(66) In some aspects, as shown in FIG. **3**, the processor **142** may instruct the vehicle user **302** to look straight (e.g., at the center of a vehicle steering wheel) at step **1** (shown as position “1” in FIG. **3**), when the processor **142** initiates the inebriation test.

(67) At step **2**, the processor **142** may activate (e.g., turn on) a first ambient vehicle light portion towards the vehicle user's left side (shown as position “2” in FIG. **3**). The processor **142** may further instruct the vehicle user **302** to track/look at the first ambient light portion “2”, without moving the head.

(68) At step **3**, the processor **142** may deactivate (turn off) the first ambient light portion and may activate a second ambient light portion that is towards the vehicle user right side (shown as location “3” in FIG. **3**). The processor **142** may again instruct the vehicle user **302** to track the second ambient light portion “3”, without moving the head.

(69) At step **4**, the processor **142** may deactivate the second ambient light portion and may activate a third ambient light portion that is towards an extreme right side (shown as location “4” in FIG. **3**). The processor **142** may repeat this process multiple times to accurately determine the inebriation test results.

(70) In other aspects, the processor **142** may illuminate/activate different visual HMI components or the vehicle panoramic display (not shown in FIG. **3**) one by one, from one side to another (in a horizontal direction), in a manner similar to the one mentioned above.

(71) In some aspects, the vehicle **300** may include an eye detector **306** (same as the second detector **156**, described in conjunction with FIG. **1**). The eye detector **306** may be configured to monitor or

capture a vehicle user eye movement **308** during the inebriation test (e.g., by tracking the vehicle user eye movement **308** from one side to another; specifically, by tracking pupil or glint trajectory). In some aspects, the processor **142** may be configured to communicate with the eye detector **306** and may receive the eye movement information from the eye detector **306**. Responsive to receiving the eye movement information, the processor **142** may compare the eye movement **308** (captured at different angles) with respective normal eye movements (information of which may be stored in the memory **118** or the memory **144**). Stated another way, the processor **142** may determine that the vehicle user eye movement **308** meets a predetermined condition. In some aspects, the predetermined condition may include twitching at extreme eye gaze or an irregular eye movement/tracking when the vehicle user's eyes track the activated vehicle visual indicators.

(72) As described above, the processor **142** may execute a single deep-learning algorithm having multi-frame analysis, deep-learning algorithms configured and/or programmed to detect eye twitching and irregular tracking, or a series of deterministic algorithms that evaluate pupil trajectory throughout the testing process. In some aspects, the algorithms may be deep or convolutional neural network architecture-based algorithms. For instance, the algorithms may include long short-term memory (LSTM) neural network based algorithms, recurrent neural network (RNN) based algorithms, spatio-temporo neural network based algorithms and/or the like.

(73) Responsive to determining twitching at extreme eye gazes or irregular eye movement, the processor **142** may determine that the vehicle user **302** may be in an inebriated state. The processor **142** may be further configured to actuate a control action(s), when the processor **142** determines that the vehicle user **302** is in the inebriated state. The control action details may be understood in conjunction with FIG. 4.

(74) In accordance with further aspects, as described above in conjunction with FIG. 1, the processor **142** may be configured to conduct the general vehicle user focus test, instead of or before the inebriation test described above. For instance, the processor **142** may conduct the general vehicle user focus test when the vehicle user **302** enters the vehicle **300** and the vehicle user **302** does not have an intoxicated operation history, when the vehicle user **302** is swerving while driving the vehicle **300** or when the vehicle user is displaying fatigue indicators.

(75) As an example, in some aspects, the processor **142** may activate the vehicle visual components (in a sequential manner) or a vehicle display system (e.g., the vehicle panoramic display), when the processor **142** conducts the general vehicle user focus test on determination of vehicle user fatigue. Responsive to visual component activation, the processor **142** may instruct the vehicle user **302** to track the activated vehicle visual components or the vehicle display system. In the general vehicle user focus test, the processor **142** may not instruct the vehicle user **302** to keep the head stationary, however the eye detector **306** may still capture the vehicle user eye movement **308**. In addition, a vehicle camera (e.g., the vehicle driver tracking camera or the vehicle occupant camera) may capture the vehicle user head rotation, as the vehicle user eyes track the activated vehicle visual components. In one aspect, the processor **142** may correlate the vehicle user head movement with the vehicle user eye movement **308**, to determine if the vehicle user **302** needs to take the inebriation test.

(76) In some aspects, the processor **142** may be configured to transmit a notification or instruct, via the infotainment system **304**, the vehicle user **302** to pull over and take some rest when the vehicle user **302** tracks the activated vehicle visual components (or the vehicle display system) properly. In other words, the processor **142** may determine that the vehicle user **302** may not be inebriated but may need some rest, when the vehicle user **302** tracks the activated vehicle visual components properly in the general vehicle user focus test. In some aspects, the processor **142** may determine that the vehicle user **302** may need rest (and may not be inebriated) when the correlation between the vehicle user head movement with the vehicle user eye movement **308** is greater than a predefined threshold. Alternatively, in cases where the vehicle user **302** is unable to track the activated vehicle visual components or the correlation is lower than a threshold, the processor **142**

may initiate the full inebriation test (as described above) to ascertain whether the vehicle user **302** is in the inebriated state.

(77) As another general vehicle user focus test example, the processor **142** may instruct the vehicle user **302** to manually switch on and/or off specific vehicle visual components (e.g., vehicle lights) in a predetermined manner, when the vehicle user **302** enters the vehicle **300**. In this case, the processor **142** may track whether the vehicle user **302** follows the instructions properly (e.g., switches on and/or off vehicle visual components as per instructions). Responsive to a determination that the vehicle user **302** follows the instruction properly, the processor **142** may not initiate the inebriation test. Alternatively, the processor **142** may initiate the inebriation test, when the processor **142** determines that the vehicle user **302** is not following the instructions.

(78) FIG. 4 illustrates an example control action(s) to prevent intoxicated operation in accordance with the present disclosure. In particular, FIG. 4 depicts a vehicle **400**, which may be operated by a vehicle user **402**. The vehicle **400** may be same as the vehicle **102**. As described above, the processor **142** may conduct an inebriation test to determine whether the vehicle user **402** is inebriated. Responsive to a determination that the vehicle user **402** is inebriated, the processor **142** may actuate an appropriate control action to prevent intoxicated operation.

(79) For instance, when the processor **142** determines that the vehicle user **402** is inebriated, the processor **142** may fetch a vehicle user contact list from the user information database **150** stored in the memory **144**. The processor **142** may be further configured to call a contact person (a mobile phone of a friend or family member) or transmit a notification to the contact person and instruct the contact person to take over the vehicle **400** as the vehicle user **402** may be inebriated (as indicated in block **404** of FIG. 4). In some aspects, the processor **142** may call the contact person via a vehicle infotainment system **406** that may be connected to a vehicle user device (not shown).

(80) In some aspects, the processor **142** may call a vehicle rideshare network/service (as indicated in block **408** of FIG. 4) or transmit a rental vehicle request/notification to a vehicle rideshare server (not shown in FIG. 4), when the vehicle user **402** is inebriated. In this case, responsive to the rental vehicle request, the vehicle rideshare service may send the rental vehicle to the vehicle **400** location, to pick up the vehicle user **402**. In one aspect, along with the rental vehicle request, the processor **142** may send a current vehicle user location and a destination vehicle user location to the vehicle rideshare server. The processor **142** may determine the current vehicle user location by using the NAV **136**. Further, the processor **142** may determine the destination vehicle user location based on the vehicle user daily activity information, stored in the user information database **150**. For example, the processor **142** may determine that the vehicle user **402** goes to home in the evening from the daily activity information. In this case, the processor **142** may determine the destination location to be vehicle user home (if the processor **142** determines the vehicle user **402** to be inebriated during evening time).

(81) In further aspects, the processor **142** may be configured to call authorities (such as police, as indicated in block **410** of FIG. 4) in certain circumstances. For instance, when the vehicle user **402** is unable to clear the inebriation test a set number of times (such as 5 times) in a specific period of time (such as 1 hour), and attempts to drive the vehicle **400**, the processor **142** may call the authorities. The processor **142** may also call the authorities, when the vehicle user **402** has an intoxicated operation history and attempts to drive the vehicle **400**. The processor **142** may further share the vehicle location to the authorities, so that the authorities may track the vehicle movement. In other aspects, a vehicle camera (not shown in FIG. 4) may capture the vehicle user's image in an inebriated state, for record purposes.

(82) In another scenario, the processor **142** may determine, via the sensory system **134**, that the vehicle user **402** is accompanied by a passenger **412** in the vehicle **400**. Responsive to the determination, the processor **142** may provide feedback/instruction, via the infotainment system **406**, to the vehicle user **402** and the passenger **412** to change the driver. In other words, the processor **142** may ask the passenger **412** to drive the vehicle **400**. In such scenarios, the processor

142 may trigger an inebriation test for the passenger **412**, to determine whether the passenger **412** is fit for driving. In this case, the processor **142** may allow the passenger **412** to drive the vehicle **400**, when the processor **142** determines that the passenger **412** is not inebriated. The processor **142** may conduct the inebriation test for the passenger **412** in the same manner as that for the vehicle user **402**.

(83) In some aspects, the processor **142** may be configured to prevent vehicle start, when the vehicle user **402** is unable to clear the inebriation test, to prevent intoxicated operation. In other aspects, the processor **142** may transmit a notification (audio or video), via the infotainment system **152**, to the vehicle user **402** informing about the intoxicated operation effects, to dissuade the vehicle user **402** from driving the vehicle **400**.

(84) In addition, when the processor **142** determines that the vehicle user **402** is fatigued (as discussed in conjunction with FIGS. **1** and **3**), the processor **142** may control the vehicle operation. For instance, the processor **142** may control vehicle speed, when the processor **142** determines that the vehicle user **402** may be fatigued, and another vehicle is close to the vehicle **400**.

(85) FIG. **5** depicts a first flow diagram of an example method **500** for providing training to a vehicle user to take an inebriation test, in accordance with the present disclosure. In particular, the method **500** describes a process to provide training to the vehicle user through vehicle audio systems, for example audio HMI components (without activating/illuminating the vehicle visual indicators). The following process is exemplary and not confined to the steps described hereafter. Moreover, alternative embodiments may include more or less steps that are shown or described herein and may include these steps in a different order than the order described in the following example embodiments.

(86) The method **500** starts at step **502**. At step **504**, the method **500** may include instructing, via the processor **142**, the vehicle user to look straight ahead. At step **506**, the method **500** may include determining, via the processor **142**, whether the vehicle user head posture is in a predetermined alignment (as discussed above). When the processor **142** determines that the head posture is not in the predetermined alignment, the method **500** moves to step **508**. At step **508**, the method **500** may include providing instructions, via the processor **142**, to the vehicle user to correct the head posture. For example, the processor **142** may transmit, via the infotainment system **152**, audio commands to tilt the head towards the center console, tilt left, and/or the like, when the vehicle user head posture is not in the predetermined alignment. Responsive to the audio command transmission at the step **508**, the method **500** may move back to the step **506**. In other words, the processor **142** may check whether the vehicle user head posture is in the predetermined alignment, when the processor **142** transmits the audio commands.

(87) In some aspects, when the processor **142** determines that the vehicle user head posture is in the predetermined alignment, the method **500** moves to step **510**. At step **510**, the method **500** may include providing instructions, via the processor **142**, to the vehicle user to hold the head and look towards a left side (such as towards a left vehicle mirror). In some aspects, the processor **142** may provide an audio instruction to the vehicle user, for example “Keep your head straight, and move your eyes to the left”.

(88) At the step **512**, the method **500** may include determining, via the processor **142**, whether the vehicle user head posture and the eye gaze are aligned, according to the instructions provided at the step **510**. If they are not aligned, then the method **500** moves back to the step **508**. If the vehicle user head posture and the eye gaze are aligned, then the processor **142** may provide an indication that they are aligned (such as by playing a chime or a confirmation audio sound), and the method **500** may move to step **514**.

(89) At step **514**, the method **500** may include providing, via the processor **142**, instructions to the vehicle user to hold the head and look towards a right side (such as towards a right vehicle mirror). At step **516**, the processor **142** may determine whether the vehicle user head posture and the eye gaze are aligned, according to the instructions provided at the step **514**. If they are not aligned, then

the method **500** may move back to the step **508**. If the vehicle user head posture and the eye gaze are aligned, then the processor **142** may provide the indication that they are aligned (such as by playing the chime or the confirmation audio sound), and the method **500** may move to step **518**. At step **518**, the method **500** may include determining, via the processor **142**, a horizontal eye nystagmus test results (as discussed above), based on the vehicle user eye movement. At step **520**, the method **500** may include performing, via the processor **142**, control action(s) based on the test results (as described in conjunction with FIG. 4). The method **500** ends at step **522**.

(90) FIG. 6 depicts a second flow diagram of an example method **600** for providing training to a vehicle user to take an inebriation test, in accordance with the present disclosure. The method **600** starts at step **602**. At step **604**, the method **600** may include instructing, via the processor **142**, the vehicle user to look straight ahead. At step **606**, the method **600** may include determining, via the processor **142**, whether the vehicle user head posture is in a predetermined alignment (as discussed above). When the processor **142** determines that the head posture is not in the predetermined alignment, the method **600** moves to step **608**. At step **608**, the method **600** may include providing instructions, via the processor **142**, to the vehicle user to correct the head posture. Responsive to providing the instructions, the method **600** may move back to the step **606**.

(91) In some aspects, when the processor **142** determines that the vehicle user head posture is in the predetermined alignment, the method **600** moves to step **610**. At step **610**, the method **600** may include activating, via the processor **142**, vehicle visual indicators to illuminate in a predetermined manner (from one side to another, in a sequential manner, as described above in conjunction with FIG. 3), and instructing the vehicle user to track the illuminated visual indicators. At step **612**, the method **600** may include detecting, via the processor **142**, vehicle user eye movement (that may be captured via the eye detector **306**, as discussed above). At step **614**, the method **600** may include determining, via the processor **142**, whether the eye movement meets a predetermined condition. In particular, the predetermined condition may include twitching at extreme eye gazes or an irregular eye movement/tracking, when the vehicle user eyes track the activated indicators.

(92) In some aspects, when the processor **142** determines that the eye movement meets the predetermined condition (e.g., when the processor **142** determines that the vehicle user may be intoxicated) at the step **614**, the method **600** moves to step **616**. At step **616**, the method **600** may include performing, via the processor **142**, control actions to prevent intoxicated operation, as discussed in conjunction with FIG. 4. Alternatively, when the processor **142** determines that the eye movement does not meet the predetermined condition at the step **614**, the method **600** moves to step **618**, at which the method **600** stops.

(93) In some aspects, the step **606** may be performed throughout the method **500**. In other words, as described above, the processor **142** may track the vehicle user head posture alignment, even when the vehicle user moves the eyes. The processor **142** may provide instructions to the vehicle user to correct head posture, as and when the vehicle user tilts the head during the method **500**.

(94) In the above disclosure, reference has been made to the accompanying drawings, which form a part hereof, which illustrate specific implementations in which the present disclosure may be practiced. It is understood that other implementations may be utilized, and structural changes may be made without departing from the scope of the present disclosure. References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a feature, structure, or characteristic is described in connection with an embodiment, one skilled in the art will recognize such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(95) Further, where appropriate, the functions described herein can be performed in one or more of hardware, software, firmware, digital components, or analog components. For example, one or

more application specific integrated circuits (ASICs) can be programmed to carry out one or more of the systems and procedures described herein. Certain terms are used throughout the description and claims to refer to particular system components. As one skilled in the art will appreciate, components may be referred to by different names. This document does not intend to distinguish between components that differ in name, but not function.

(96) It should also be understood that the word “example” as used herein is intended to be non-exclusionary and non-limiting in nature. More particularly, the word “example” as used herein indicates one among several examples, and it should be understood that no undue emphasis or preference is being directed to the particular example being described.

(97) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Computing devices may include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above and stored on a computer-readable medium.

(98) With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating various embodiments and should in no way be construed so as to limit the claims.

(99) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

(100) All terms used in the claims are intended to be given their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Claims

1. A method to prevent intoxicated vehicle operation, the method comprising: obtaining, via a processor, vehicle user body portion posture from a first vehicle detector; determining, via the processor, whether the vehicle user body portion posture is in a predefined alignment; activating, via the processor, a plurality of vehicle visual indicators to illuminate in a predefined manner, responsive to the determination that the vehicle user body portion posture is in the predefined alignment; generating, via the processor, an instruction to move vehicle user eyes; obtaining, via

the processor, a vehicle user eye movement information based on the instruction, from a second vehicle detector; determining, via the processor, based on the user eye movement information, that a vehicle user meets a predetermined condition, wherein the predetermined condition is associated with a vehicle user inebriated state; and actuating, via the processor, a control action responsive to the determination that the user eye movement meets the predetermined condition.

2. The method of claim 1 further comprising: providing, via the processor, output instructing the vehicle user to correct the vehicle user body portion posture, based on a determination that the vehicle user body portion posture is not in the predefined alignment; and obtaining, via the processor, the vehicle user body portion posture responsive to providing the output.

3. The method of claim 1, wherein the first vehicle detector is configured to detect a vehicle user head posture.

4. The method of claim 1, wherein the plurality of vehicle visual indicators comprises Human-Machine Interface (HMI) visual components or a vehicle display.

5. The method of claim 1, wherein the predetermined condition comprises twitching at extreme eye gazes or an irregular eye tracking when the vehicle user eyes track the plurality of vehicle visual indicators.

6. The method of claim 1, wherein activating the plurality of vehicle visual indicators to illuminate in the predefined manner comprises illuminating the plurality of vehicle visual indicators in a sequential manner and in a horizontal direction.

7. The method of claim 1, wherein the control action comprises at least one of: disabling a vehicle start, transmitting a vehicle user inebriated state notification to an acquaintance user device or a vehicle ridesharing server, and providing an instruction to a non-inebriated vehicle passenger to drive a vehicle.

8. A vehicle for preventing intoxicated operation, the vehicle comprising: a first vehicle detector configured to detect a vehicle user body portion posture; a second vehicle detector configured to detect a vehicle user eye movement; and a processor communicatively coupled to the first vehicle detector and the second vehicle detector, wherein the processor is configured to: provide instructions to a vehicle user to position the vehicle user body portion posture in a predefined alignment; obtain the vehicle user body portion posture from the first vehicle detector; determine whether the vehicle user body portion posture is in the predefined alignment; activate a plurality of vehicle visual indicators to illuminate in a predefined manner, responsive to the determination that the vehicle user body portion posture is in the predefined alignment; generate an instruction to move vehicle user eyes; obtain a vehicle user eye movement information based on the instructions, from the second vehicle detector; determine, based on the user eye movement information, that a user meets a predetermined condition, wherein the predetermined condition is associated with a vehicle user inebriated state; and actuate a control action responsive to the determination that the vehicle user eye movement meets the predetermined condition.

9. The vehicle of claim 8, wherein the processor is further configured to: generate a second instruction to correct the vehicle user body portion posture, based on a determination that the vehicle user body portion posture is not in the predefined alignment; and obtain the vehicle user body portion posture responsive to providing the instruction.

10. The vehicle of claim 8, wherein the first vehicle detector is configured to detect a vehicle user head posture.

11. The vehicle of claim 8, wherein the plurality of vehicle visual indicators comprises Human-Machine Interface (HMI) visual components or a vehicle display.

12. The vehicle of claim 8, wherein the predetermined condition comprises twitching at extreme eye gazes or an irregular eye tracking when the vehicle user eyes track the plurality of vehicle visual indicators.

13. The vehicle of claim 8, wherein the processor activates the plurality of vehicle visual indicators by illuminating the plurality of vehicle visual indicators in a sequential manner and in a horizontal

direction.

14. The vehicle of claim 8, wherein the control action comprises at least one of: disabling a vehicle start, transmitting a vehicle user inebriated state notification to an acquaintance user device or a vehicle ridesharing server, and providing an instruction to a non-inebriated vehicle passenger to drive a vehicle.

15. A non-transitory computer-readable storage medium having instructions stored thereupon which, when executed by a processor, cause the processor to: provide instructions to a vehicle user to position a vehicle user body portion posture in a predefined alignment; obtain the vehicle user body portion posture from a first vehicle detector; determine whether the vehicle user body portion posture is in the predefined alignment; activate a plurality of vehicle visual indicators to illuminate in a predefined manner, responsive to the determination that the vehicle user body portion posture is in the predefined alignment; generate an instruction to move vehicle user eyes to track the plurality of vehicle visual indicators; obtain a vehicle user eye movement information based on the instructions, from a second vehicle detector; determine, based on the vehicle eye movement information, that a vehicle user eye movement meets a predetermined condition, wherein the predetermined condition is associated with a vehicle user inebriated state; and actuate a control action responsive to the determination that the vehicle user eye movement meets the predetermined condition.

16. The non-transitory computer-readable storage medium of claim 15, further comprising: providing, via the processor, output instructing the vehicle user to correct the vehicle user body portion posture, based on a determination that the vehicle user body portion posture is not in the predefined alignment; and obtaining, via the processor, the vehicle user body portion posture responsive to providing the output.

17. The non-transitory computer-readable storage medium of claim 15, wherein the first vehicle detector is configured to detect a vehicle user head posture.

18. The non-transitory computer-readable storage medium of claim 15, wherein the plurality of vehicle visual indicators comprises Human-Machine Interface (HMI) visual components or a vehicle display.

19. The non-transitory computer-readable storage medium of claim 15, wherein the predetermined condition comprises twitching at extreme eye gazes or an irregular eye tracking when the vehicle user eyes track the plurality of vehicle visual indicators.

20. The non-transitory computer-readable storage medium of claim 15, wherein activating the plurality of vehicle visual indicators to illuminate in the predefined manner comprises illuminating the plurality of vehicle visual indicators in a sequential manner and in a horizontal direction.
